Deploy Rancher on NetApp HCI

HCI

Ann-Marie Grissino, Michael Wallis, Aksel Davis, Dave Bagwell
May 26, 2021

This PDF was generated from https://docs.netapp.com/us-en/hci/docs/task_rancher_deploy.html on July 16, 2021. Always check docs.netapp.com for the latest.

Table of Contents

Deploy Rancher on NetApp HCI . . . 1
What happens when you deploy Rancher on NetApp HCI? . . . 1
Steps to deploy Rancher on NetApp HCI . . . 1
What’s next? . . . 4

Deploy Rancher on NetApp HCI

To use Rancher on your NetApp HCI environment, you first deploy Rancher on NetApp HCI.

Before starting the deployment, be sure to check the datastore free space and other requirements for Rancher on NetApp HCI.

Rancher Support is not included in your NetApp Support Edge agreement. Contact NetApp Sales or your reseller for options. If you purchase Rancher Support from NetApp, you will receive an email with instructions.

What happens when you deploy Rancher on NetApp HCI?

The deployment involves the following steps, each described further:

- Use the NetApp Hybrid Cloud Control to initiate the deployment.
- The Rancher deployment creates a management cluster, which includes three virtual machines.

  Each virtual machine is assigned all Kubernetes roles for both the Control Plane and Worker. This means that the Rancher UI is available on each node.

- The Rancher Control Plane (or *Rancher Server*) is also installed, using the NetApp HCI node template in Rancher for easier deployment. The Rancher Control Plane automatically works with the configuration used in the NetApp Deployment Engine, which was used to build the NetApp HCI infrastructure.
- After deployment, you will receive an email from NetApp providing you with the option to register for NetApp Support on Rancher deployments on NetApp HCI.
- After deployment, Dev and Ops teams can then deploy their user clusters, similar to any Rancher deployment.

Steps to deploy Rancher on NetApp HCI

- Access the NetApp Hybrid Cloud Control
- Deploy Rancher on NetApp HCI
- Verify your deployment by using vCenter Server

Access the NetApp Hybrid Cloud Control

To begin the deployment, access the NetApp Hybrid Cloud Control.

1. Open a web browser and browse to the IP address of the management node. For example:

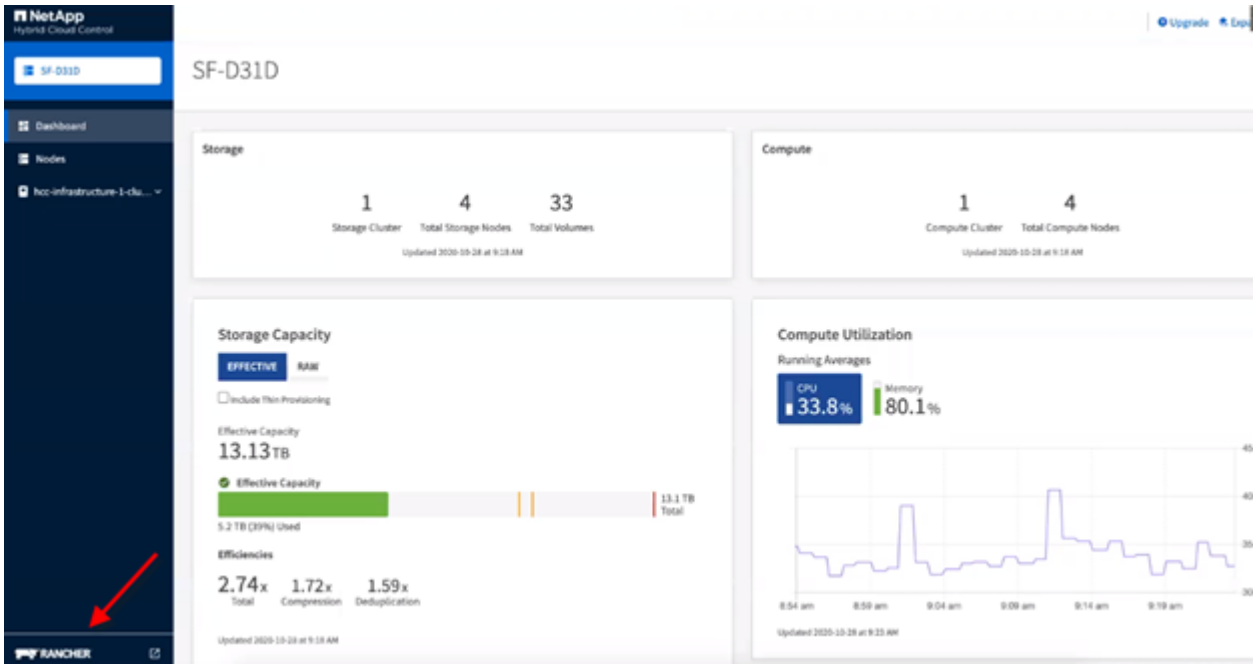
   https://<*ManagementNodeIP*>

2. Log in to NetApp Hybrid Cloud Control by providing the NetApp HCI storage cluster administrator credentials.

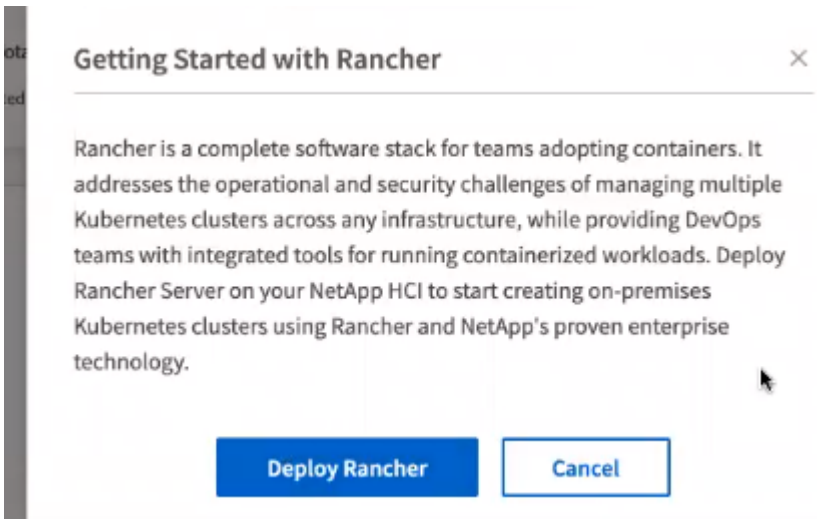
   The NetApp Hybrid Cloud Control interface appears.

Deploy Rancher on NetApp HCI

1. From the Hybrid Cloud Control, click the **Rancher** icon in the lower left of the navigation bar.

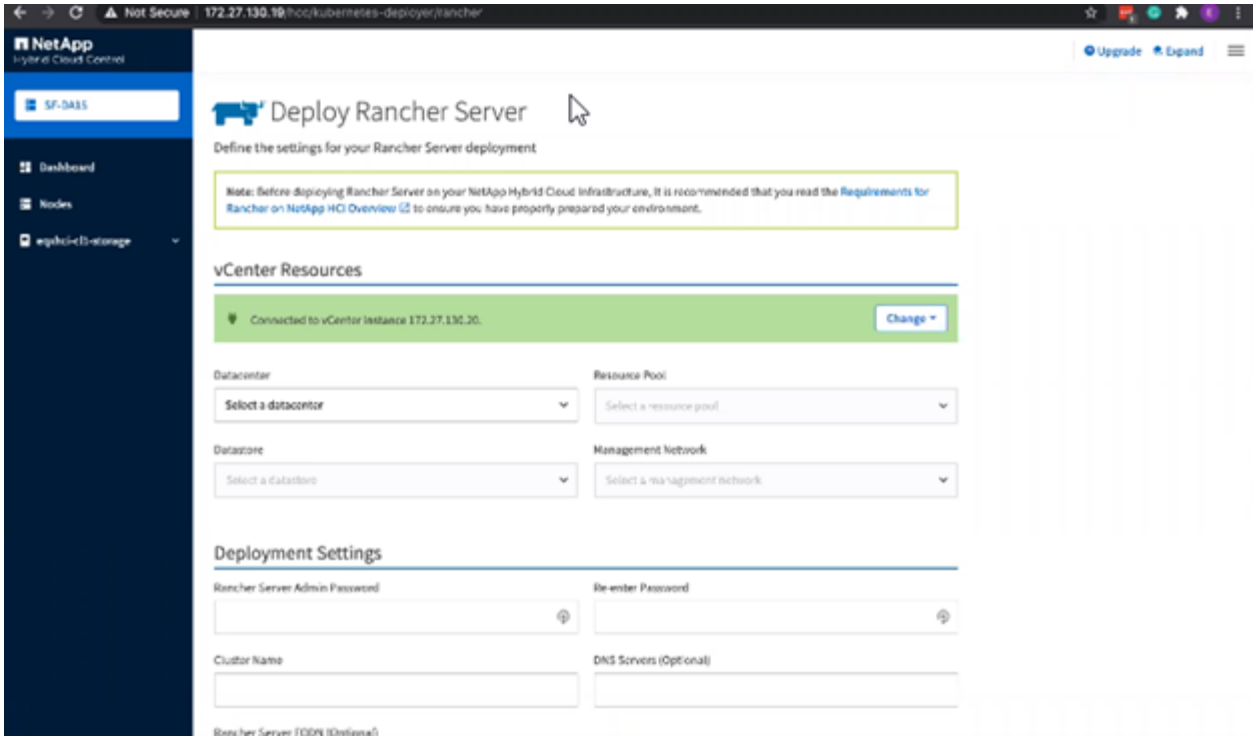


A popup window shows a message about getting started with Rancher.



2. Click **Deploy Rancher**.

   The Rancher UI appears.



Your vCenter credentials are collected based on your NetApp Deployment Engine installation.

3. Enter **vCenter Resources** information. Some fields are described next.
   - **Datacenter**: Select a datacenter. After you select the datacenter, all other fields are prepopulated, although you can change them.
   - **Datastore**: Select a datastore on the NetApp HCI storage nodes. This datastore should be resilient and accessible to all of the VMware hosts. Do not select a local datastore that is accessible to only one of the hosts.
   - **Management network**: This should be accessible from the management stations and from the virtual machine network where the user clusters will be hosted.
4. Enter **Deployment Settings** information:
   - **DNS Servers**: Optional. If you use load balancing, enter the internal DNS server information.
   - **Rancher Server FQDN**: To ensure that the Rancher Server remains available during node failures, provide a fully-qualified domain name (FQDN) that your DNS server can resolve to any of the IP addresses assigned to the Rancher Server cluster's nodes. This FQDN with the "https" prefix becomes the Rancher URL that you will use to access your Rancher implementation.

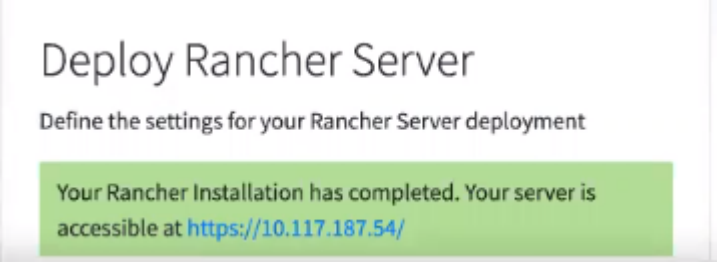
     If no domain name is provided, wildcard DNS will be used instead and you will be able to access the Rancher Server using one of the URLs presented after the deployment completes.
5. Enter **Advanced Settings** information:
   - **Assign Static IP Addresses**: If you enable static IP addressing, provide starting IP addresses for three IPv4 addresses in sequence, one for each management cluster virtual machine. Rancher on NetApp HCI deploys three management cluster virtual machines.
   - **Configure Proxy Server**:
6. Review and select the checkbox for the Rancher End User License Agreement.
7. Review and select the checkbox to acknowledge information about Rancher software.

8. Click **Deploy**.

   A bar indicates the deployment progress.

   The Rancher deployment could take about 15 minutes.

   When the deployment is complete, Rancher displays a message about the completion and provides a Rancher URL.



9. Record that Rancher URL that Sdisplays at the end of the deployment. You will use this URL to access the Rancher UI.

Verify your deployment by using vCenter Server

In your vSphere client, you can see the Rancher management cluster, which includes the three virtual machines.

Once you have finished deployment, do not modify the configuration of the Rancher server virtual machine cluster or remove the virtual machines. Rancher on NetApp HCI relies on the deployed RKE management cluster configuration to function normally.

What's next?

After deployment, you can do the following:

- Complete post-deployment tasks
- Install Trident with Rancher on NetApp HCI
- Deploy user clusters and applications
- Manage Rancher on NetApp HCI
- Monitor Rancher on NetApp HCI

Find more information

- Rancher deployment troubleshooting
- Rancher documentation about architecture
- Kubernetes terminology for Rancher
- NetApp HCI Resources page

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.